

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 26)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

For each of the following investigations, name the most appropriate statistical test.

[4 marks]

Whether noise level decreases as distance from a road increases

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Whether the numbers of dormice differ between woodlands with different management ..

.....

Whether mean dissolved oxygen differs between two streams

.....

Whether crop yield increases with fertiliser application (ranked data)

.....

0 1 . 2

Explain why the chi-squared test cannot be used for data such as wind speed or plant height.

[1 mark]

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.....

0 1 . 3

A Spearman's rank test on 12 pairs of data gave $r_s = -0.72$. The critical value at $p = 0.05$ for $n = 12$ is 0.59. Explain what can be concluded.

[3 marks]

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0 1 . 4

Explain the purpose of error bars on a graph of mean values.

[2 marks]

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10

0 2 Table 1 shows information about some greenhouse gases.

Table 1

Gas	Main anthropogenic sources	Approximate residence time	Global warming potential (per molecule, relative to CO ₂ over 100 years)
Carbon dioxide	Fossil fuel combustion, deforestation, cement	100+ years (variable)	1
Methane	Rice paddies, cattle, landfill, gas leaks, coal mines	About 12 years	28
Nitrous oxide (NO _x)	Fertilisers, combustion	About 120 years	265
CFCs	Refrigerants, aerosols (now banned)	50-100+ years	5 000-10 000
Tropospheric ozone	Secondary pollutant from NO _x and hydrocarbons	Days-weeks	-

0 2 . **1** Explain the natural greenhouse effect.

[3 marks]

0 2 . **2** Use Table 1 to explain why carbon dioxide is considered the most important anthropogenic greenhouse gas even though methane and CFCs absorb far more infrared per molecule.

[3 marks]

0 2 . 3

Explain why reducing methane emissions would produce a faster reduction in warming than reducing CO₂ emissions by the same mass.

[2 marks]

0 2 . 4

Give **two** methods of reducing methane emissions from agriculture or waste.

[2 marks]

1

2

10

0 3

A student measured the mass of carbon stored per hectare in three land uses in the same region.

Table 1

Land use	Carbon in vegetation / t ha ⁻¹	Carbon in soil dead organic matter / t ha ⁻¹
Mature woodland	120	95
Permanent pasture	3	80
Arable (ploughed annually)	2	45

0 3 . 1

Calculate the total carbon lost per hectare when mature woodland is converted to arable farmland.

[1 mark]

Answer t/ha

03 . 2

Explain why the soil carbon in arable land is so much lower than in pasture, even though the vegetation carbon is similar.

[2 marks]

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.....

03 . 3

Suggest **two** ways the arable farmer could increase soil carbon storage.

[2 marks]

1

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2

.....

5

04 . 1

Which of the following best describes the **precautionary principle**? Tick **one** box.

[1 mark]

- A A waste is assumed to cause pollution until research shows it is unlikely to do so ☒
- B The polluter must pay for any damage caused ☐
- C Emissions should be as low as reasonably achievable ☐
- D Only pollutants known to be toxic need to be controlled ☐

04 . 2

Explain why the polluter pays principle works best when penalties are high and enforcement is reliable.

[2 marks]

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.....

.....

 0 4 . 3 Give **two** choices about emission location that reduce the severity of pollution. **[2 marks]**

1

2

5

0 5 In 2020 the tanker MV Wakashio ran aground on a coral reef off Mauritius and leaked about 1000 tonnes of fuel oil into a shallow lagoon near mangroves and a marine park.

0 5 . 1 Explain why inflatable booms were deployed immediately, and give **one** limitation of booms. **[3 marks]**

0 5 . 2 Describe how skimmers and absorbent materials remove oil, and state how the collected oil is disposed of. **[3 marks]**

0 5 . 3

Explain why dispersants were **not** used in the lagoon.**[2 marks]**

0 5 . 4

Explain how bioremediation could help remove the oil that remained on the mangrove sediments, and suggest how it could be accelerated.

[2 marks]**10**

0 6

Table 1 shows the mean surface temperature of the Earth with and without an atmosphere containing greenhouse gases.

Table 1

Condition	Mean surface temperature / °C
No greenhouse effect	-18
Natural greenhouse effect	15

0 6 . 1

Use Table 1 to calculate the warming produced by the natural greenhouse effect.

[1 mark]

Answer

&deg;C

0 6 . 2

Explain how the natural greenhouse effect warms the Earth's surface.

[4 marks]

06 . 3

Explain what is meant by a **greenhouse gas**.

[1 mark]

06 . 4

The wavelengths of radiation arriving at the Earth are different from those leaving. Describe this difference and explain why it matters to the energy budget.

[2 marks]

06 . 5

Explain why a change in the movement of energy through the atmosphere could affect the distribution of species.

[2 marks]

10

07

Table 1 shows the volume of water in some reservoirs of the hydrological cycle and the annual transfer rates through them.

Table 1

Reservoir	Volume / $\times 10^{15}$ dm^3	Annual transfer (inflow = outflow) / $\times 10^{15} \text{ dm}^3 \text{ yr}^{-1}$	Percentage of global water / %
Oceans	1 340 000	425	97
Ice caps and glaciers	24 000	3	1.7
Groundwater	23 000	12	1.7
Lakes and rivers	180	40	0.013
Soil moisture	17	80	0.001
Atmosphere	13	480	0.001

07**1**

The residence time of water in a reservoir is given by: residence time = volume in reservoir / annual transfer rate. Use Table 1 to calculate the residence time of water in the atmosphere. Give your answer to the nearest whole day.

[3 marks]**Answer**

days

07**2**

Calculate the residence time of water in the oceans.

[1 mark]**Answer**

years

07**3**

Explain why the atmosphere is an important reservoir even though it holds only 0.001% of global water.

[2 marks]

07 . 4

Explain why human changes to a small reservoir such as soil moisture are more obvious than changes to the oceans.

[2 marks]

07 . 5

Explain why the hydrological cycle is described as being in dynamic equilibrium.

[2 marks]

10

08 . 1

Explain how each of the following human activities increases soil erosion.

[4 marks]

Overgrazing

Reducing soil biota

08 . 2

Explain why landslides become more likely on steep slopes some years after deforestation.

[2 marks]

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08 . 3

Describe **four** environmental impacts of accelerated soil erosion away from the field where it occurs.

[4 marks]

1

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2

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3

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4

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10

09 . 1

Explain the difference between a **mutagen**, a **carcinogen** and a **teratogen**.

[3 marks]

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09 . 2

Explain why a birth abnormality caused by a teratogen cannot be inherited by future generations, whereas one caused by a mutagen may be.

[2 marks]

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09 . 3

Distinguish between **somatic** and **gonadic** effects of a mutagen.

[2 marks]

09 . 4

Explain what is meant by **synergism** and give an example.

[3 marks]

10

10

Table 1 shows the mean daily solar energy received on a horizontal surface at three locations.

Table 1

Location	Latitude	Mean solar energy in June / kWh m ⁻² day ⁻¹	Mean solar energy in December / kWh m ⁻² day ⁻¹
Sahara desert	25°N	7.8	4.2
Southern England	51°N	5.0	0.7
Northern Norway	69°N	4.6	0.0

10 . 1

Use Table 1 to calculate how many times more solar energy is received in December in the Sahara than in southern England.

[1 mark]

Answer times

1 0 . 2 Explain why northern Norway receives more solar energy in June than in December. **[2 marks]**

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1 0 . 3 Explain why concentrating solar power systems using parabolic reflectors only work where there is no cloud. **[2 marks]**

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1 0 . 4 Explain why the low energy density of solar power is a problem for large-scale generation. **[1 mark]**

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1 0 . 5 Evaluate the extent to which new technologies could overcome the problems of using solar power as a large-scale energy resource. **[9 marks]**

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[illegible]

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15

Write an essay on one of the following topics.

$$\boxed{1}\boxed{1}.\boxed{1}$$

Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment.

[25 marks]

OR

1 1 . 2

Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

END OF QUESTIONS